

# Toric Varieties

Labix

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**Abstract**

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# 1 Introduction to Toric Varieties

## 1.1 Characters and One Parameter Subgroups

### Definition 1.1.1 (The Group of One Parameter Subgroups)

Let  $k$  be a field. Let  $T$  be a torus over  $k$ . Define the group of one parameter subgroups of  $T$  to be the set

$$\lambda(T) = \{ \lambda : \mathbb{G}_m \rightarrow T \mid \lambda \text{ is a one-parameter subgroup} \}$$

together with pointwise multiplication.

### Lemma 1.1.2

Let  $k$  be a field. Let  $T$  be a split torus over  $k$ . Then the map

$$\langle -, - \rangle : \chi(T) \times \lambda(T) \rightarrow \chi(\mathbb{G}_m) \cong \mathbb{Z}$$

given by  $\langle \chi, \lambda \rangle \mapsto \chi \circ \lambda$  is a perfect pairing.

Fix a split torus  $T$ . We have constructed the group of characters  $\chi(T)$  and the group of one parameter subgroups  $\lambda(T)$ .

Now start with a lattice  $M \cong \mathbb{Z}^n$ . We think of it as specifying a group of characters. We may recover the underlying split torus by  $T = \text{Spec}(k[M])$ . The group one parameter subgroups is deduced by taking the dual  $\lambda(T) = \text{Hom}_{\mathbf{Ab}}(M, \mathbb{Z})$ .

Finally, start with a lattice  $N \cong \mathbb{Z}^n$ . We think of it as specifying a group of one parameter subgroups. We may recover the underlying split torus by  $T = N \otimes_{\mathbb{Z}} k^*$ . The group of characters can be recovered by taking the dual  $\chi(T) = \text{Hom}_{\mathbf{Ab}}(N, \mathbb{Z})$ .

Thus any one of  $T, M, N$  is enough data to determine a torus.

## 1.2 General Toric Varieties

### Definition 1.2.1 (Toric Varieties)

Let  $X$  be an irreducible variety over  $\mathbb{C}$ . We say that  $X$  is a toric variety if the following are true.

- $X$  contains a torus  $T_X$  as a Zariski open subset.
- The action of  $T_X$  on itself extends to  $X$ .

The idea is that toric varieties consists of the underlying points together with limit points. Recall that there is a vector space decomposition

$$\mathbb{C}[T] \cong \bigoplus_{\chi \in \chi(T)} \mathbb{C} \cdot \chi$$

This means that the character functions span coordinate ring of the torus as a vector space.

The one parameter subgroups  $\lambda \in \lambda(T)$  traces out the curves on the torus. To add in limit points to the torus  $T$  is the same as adding in the limit  $\lim_{t \rightarrow 0} \lambda(t)$  since  $\lambda$  is defined on  $\mathbb{C} \setminus \{0\}$ .

For example, suppose we wish to add in the point  $(1, 0) \in \mathbb{C}^2$  to the torus  $T = (\mathbb{C}^\times)^2$ . When we add in limit points, we expect certain characters on the torus to no longer be elements of the coordinate ring of the toric variety. Let  $x^a y^b \in \mathbb{C}[T]$  be a general character of a torus. Then it is well defined on the point  $(1, 0)$  if and only if  $b > 0$ . This adds a linear inequality to the lattice  $\chi(T)$ . On the other hand, we may think of  $(1, 0)$  as the limit point of the one parameter subgroup  $\lambda(t) = (t^1, t^0)$ . However we also need to consider the equivariant action of the torus, so adding just the point  $(1, 0)$  is not enough.

**Example 1.2.2**

The following are toric varieties.

- The torus  $\mathbb{G}_m^n(\mathbb{C})$ .
- Affine space  $\mathbb{A}_{\mathbb{C}}^n$ .
- Projective space  $\mathbb{P}_{\mathbb{C}}^n$ .

The equivariant structure of the torus encodes a lot of the algebraic geometry into combinatorial data or more manageable structure. A lot of common varieties are toric, or are subvarieties of complete intersections of toric varieties.

There are two ways to construct normal toric varieties:

(1): Make an open cover of it via affine schemes that each corresponds to some polyhedral cone, then glue the open cover from combinatorial data.

(2): Encode a projective variety as the proj construction given by a polytope.

**Definition 1.2.3 (Torus Factors)**

Let  $X$  be a toric variety. We say that  $X$  has a torus factor if there is an equivariant isomorphism

$$X \cong Y \times T^k$$

for some toric variety  $Y$ .

**Definition 1.2.4 (Toric Morphisms)**

Let  $X, Y$  be toric varieties over  $\mathbb{C}$ . Let  $\phi : XY$  be an equivariant morphism. We say that  $\phi$  is a toric morphism if the following are true.

- $\phi(T_X) \subseteq T_Y$ .
- $\phi|_{T_X} : T_X \rightarrow T_Y$  is an equivariant morphism.

### 1.3 Equivalent Characterizations of Affine Toric Varieties

**Example 1.3.1**

The following are affine toric varieties.

- The cuspidal cubic  $\mathbb{V}(x^3 - y^2)$ .
- The variety  $\mathbb{V}(xy - zw)$ .

**Proof**

- Notice that the map  $\mathbb{C}^\times \rightarrow \mathbb{V}(x^3 - y^2)$  given by  $t \mapsto (t^2, t^3)$  defines a dense torus whose only point outside the torus is the origin. Hence the action of the torus extends naturally to all of the variety.
- Similarly, the map  $(\mathbb{C}^\times)^3 \rightarrow \mathbb{V}(xy - zw)$  given by  $(t_1, t_2, t_3) \mapsto (t_1, t_2, t_3, t_1 t_2 t_3^{-1})$  defines a torus inside the variety. The remaining points outside the torus is given by  $\mathbb{V}(zw) \cup \mathbb{V}(xy)$  the torus action extends naturally along the four axes.

■

There are a few ways to define affine toric varieties. One such way is to choose directions in the torus to take closure.

**Definition 1.3.2 (Affine Toric Varieties Associated to Characters)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Define the affine toric

variety associated to  $\mathcal{A}$  to be

$$Y_{\mathcal{A}} = \overline{\text{im}(\Phi_{\mathcal{A}})}$$

where  $\Phi_{\mathcal{A}} : T \rightarrow \mathbb{A}_{\mathbb{C}}^s$  is the map defined by  $\Phi_{\mathcal{A}}(t) = (\chi_1(t), \dots, \chi_s(t))$

Here we think of a choice of basis for  $\chi(T)$  is a choice integral coordinates on  $T$ . Then choosing  $\mathcal{A}$  is to choose which rays in the  $\chi(T)$  to take closure.

### Proposition 1.3.3

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Then the following are true.

- $Y_{\mathcal{A}}$  is an affine toric variety.
- $\chi(T_{Y_{\mathcal{A}}}) = \langle \mathcal{A} \rangle$ .
- $\dim(Y_{\mathcal{A}}) = \text{rank}(\langle \mathcal{A} \rangle)$ .
- The ideal of  $Y_{\mathcal{A}}$  is given by

$$\mathbb{I}(Y_{\mathcal{A}}) = I_{\ker(\phi)} = \langle x^{\alpha} - x^{\beta} \mid \alpha, \beta \in \mathbb{N}^s, \alpha - \beta \in \ker(\phi) \rangle$$

where  $\phi : \mathbb{Z}^s \rightarrow \chi(T)$  is the group homomorphism defined by  $e_i \mapsto \chi_i$ .

### Proof

Notice that the image of  $\Phi_{\mathcal{A}}$  lands in  $\mathbb{G}_m^s$ . Hence  $\Phi_{\mathcal{A}}$  is a map of torii. By lmm we know that  $\text{im}(\Phi_{\mathcal{A}})$  is a closed torus in  $\mathbb{G}_m^s$ . By the relative closure formula, we have that

$$Y_{\mathcal{A}} \cap \mathbb{G}_m^s = \text{im}(\Phi_{\mathcal{A}})$$

Then by definition of subspace topology on  $Y_{\mathcal{A}}$ , we conclude that  $\text{im}(\Phi_{\mathcal{A}})$  is an open subset of  $Y_{\mathcal{A}}$ . This shows that  $Y_{\mathcal{A}}$  contains a torus as an open subset. Also, since  $\text{im}(\Phi_{\mathcal{A}})$  is irreducible and is an open subset of  $Y_{\mathcal{A}}$ , we conclude that  $Y_{\mathcal{A}}$  is irreducible.

Since  $T$  acts on itself for being a torus, the map  $\Phi_{\mathcal{A}}$  gives the action of  $\text{im}(\Phi_{\mathcal{A}})$  on  $\mathbb{G}_m^s$  that sends varieties to varieties. Let  $t \in \text{im}(\Phi_{\mathcal{A}})$ . Since  $\text{im}(\Phi_{\mathcal{A}})$  is a torus, we have that

$$\text{im}(\Phi_{\mathcal{A}}) = t \cdot \text{im}(\Phi_{\mathcal{A}}) \subseteq t \cdot Y_{\mathcal{A}}$$

By definition of closure, we must have  $Y_{\mathcal{A}} \subseteq t \cdot Y_{\mathcal{A}}$ . Replacing  $t$  with  $t^{-1}$  show that  $t \cdot Y_{\mathcal{A}} \subseteq Y_{\mathcal{A}}$ . Hence the action of  $\text{im}(\Phi_{\mathcal{A}})$  gives an action on  $Y_{\mathcal{A}}$ . Thus  $Y_{\mathcal{A}}$  is an affine toric variety. ■

Both the toric varieties  $\mathbb{V}(x^3 - y^2)$  and  $\mathbb{V}(xy - zw)$  can be defined by first writing down the torus in this way.

### Definition 1.3.4 (Toric Ideals)

Let  $I \subseteq \mathbb{C}[x_1, \dots, x_n]$  be an ideal. We say that  $I$  is a toric ideal if the following are true.

- $I$  is a lattice ideal.
- $I$  is a prime ideal.

### Lemma 1.3.5

Let  $I \subseteq \mathbb{C}[x_1, \dots, x_n]$  be an ideal. Then  $I$  is toric if and only if it is prime and is generated by binomials of the form  $x_{i_1}^{\alpha_1} \cdots x_{i_n}^{\alpha_n} - x_{j_1}^{\beta_1} \cdots x_{j_m}^{\beta_m}$ .

### Proof

If  $I$  is a toric ideal then it is clearly prime and generated by binomials. So suppose that  $I$  is prime and is generated by binomials of the above form. Notice that  $\mathbb{V}(I) \cap (\mathbb{C}^{\times})^n$  is non-empty since it contains  $(1, \dots, 1)$ . We check that it is a subgroup of  $(\mathbb{C}^{\times})^n$ . Let  $u, v \in \mathbb{V}(I) \cap (\mathbb{C}^{\times})^n$ . Then we know that  $u^{\alpha} - u^{\beta} = 0 = v^{\alpha} - v^{\beta}$ . Then we have

$$(uv)^{\alpha} - (uv)^{\beta} = u^{\alpha}v^{\alpha} - u^{\beta}v^{\alpha} + u^{\beta}v^{\alpha} - u^{\beta}v^{\beta} = (u^{\alpha} - u^{\beta})v^{\alpha} + u^{\beta}(v^{\alpha} - v^{\beta}) = 0$$

Hence products are closed. Also, we have Hence inverses are closed. Thus  $\mathbb{V}(I) \cap (\mathbb{C}^{\times})^n$  is a

subgroup of  $(\mathbb{C}^\times)^n$ . By Imm, we conclude that  $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$  is a torus. For each  $1 \leq i \leq n$ , projecting to the  $i$ th coordinate gives a character  $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n \rightarrow (\mathbb{C}^\times)^n \rightarrow \mathbb{C}^\times$ . Let  $\mathcal{A}$  be the set of these characters. Then by construction we have  $\mathbb{V}(I) = Y_{\mathcal{A}}$ . By the Nullstellensatz, we have  $I = \mathbb{I}(\mathbb{V}(I)) = \mathbb{I}(Y_{\mathcal{A}})$ . Hence by the above we conclude that  $I$  is toric. ■

### Proposition 1.3.6

Let  $S$  be an affine semigroup. Then  $\text{Spec}(\mathbb{C}[S])$  is an affine toric variety.

Fix a torus  $T$ . For each  $\chi \in \chi(T)$ , notice that if  $X$  is a variety such that  $T \subseteq X$  is dense, then  $\chi \in \mathbb{C}(X)$ . Then specifying an affine semigroup  $S \subseteq \chi(T)$  is the same as choosing which characters you want to be regular functions instead of rational functions.

### Proposition 1.3.7

Let  $X$  be an irreducible affine variety over  $\mathbb{C}$ . Then the following are equivalent.

- $X$  is a affine toric variety.
- There exists a torus  $T$  and a finite subset  $\mathcal{A} \subseteq \chi(T)$  such that  $X = Y_{\mathcal{A}}$ .
- There exists a toric ideal  $I$  of  $\mathbb{C}[x_1, \dots, x_n]$  such that  $X \cong \text{Spec}\left(\frac{\mathbb{C}[x_1, \dots, x_n]}{I}\right)$ .
- There exists an affine semigroup  $S$  such that  $X \cong \text{Spec}(\mathbb{C}[S])$ .

### Example 1.3.8 (The Rational Normal Cone)

Let  $d \in \mathbb{N}^+$ . Let  $\phi : \mathbb{C}^2 \rightarrow \mathbb{C}^{d+1}$  be the morphism defined by

$$\phi(s, t) = (s^d, s^{d-1}t, \dots, st^{d-1}, t^d)$$

Define the rational normal cone of degree  $d$  to be

$$\widehat{C}_d = \text{im}(\phi)$$

Then the following are true.

- $\mathbb{I}(\widehat{C}_d) = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d)$ .
- $\widehat{C}_d$  is an affine toric variety.

### Proof

Let  $I = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d) \subseteq \mathbb{C}[x_1, \dots, x_{d+1}]$ . Clearly we have  $\widehat{C}_d \subseteq \mathbb{V}(I)$ . For the reverse inclusion, notice first that if  $d = 0$  then the result is trivial. Now let  $p = (p_1, \dots, p_{d+1}) \in \mathbb{V}(I)$ . If  $p = 0$  then clearly  $p \in \text{im}(\phi)$ . Now let  $m = \min\{k \mid p_k \neq 0\}$ . I claim that  $m = 1$  or  $d + 1$ . Notice that we have  $x_{k-1}x_{k+1} - x_k^2 \in I$  for  $2 \leq k \leq d$ . If  $2 \leq m \leq d$ , then  $p_{m-1} = 0$  so that  $p_m = 0$ , contradicting the minimality of  $m$ . Hence  $m = 1$  or  $d + 1$ . If  $m = d + 1$ , then  $p = (0, \dots, 0, p_{d+1})$ . Since  $p_{d+1} \in \mathbb{C}^\times$ , choose  $t \in \mathbb{C}$  such that  $t^d = p_{d+1}$ . Then we have that  $(0, t) \mapsto p$  so  $p \in \text{im}(\phi)$ . If  $m = 1$ , let  $r = p_2/p_1$ . Using the relation  $x_1 x_{j+1} = x_2 x_j$  we deduce that  $p_{j+1}/p_j = r$ . Hence  $p$  is given by  $p = (p_1, p_1 r, \dots, p_1 r^d)$ . Since  $p_1 \in \mathbb{C}^\times$ , choose  $s \in \mathbb{C}$  such that  $s^d = p_1$ . Then we have  $(s, rs) \mapsto p$  so that  $p \in \widehat{C}_d$ . Thus we are done. ■

### Example 1.3.9

Let  $T = (\mathbb{C}^\times)^3$ . Consider the toric variety  $X = \mathbb{V}(xy - zw)$ . Then the following are true.

- Let  $\mathcal{A} = \{\chi_1(t_1, t_2, t_3) = t_1, \chi_2(t_1, t_2, t_3) = t_2, \chi_3(t_1, t_2, t_3) = t_3, \chi_4(t_1, t_2, t_3) = t_1 t_2 t_3^{-1}\} \subseteq \chi(T)$ . Then  $Y_{\mathcal{A}} = X$ .
- The toric ideal of  $X$  is given by  $(xy - zw)$ .
- Consider the isomorphism  $\chi(T) \cong \mathbb{Z}^3$  given by  $\chi_i \mapsto e_i$  for  $1 \leq i \leq 3$ . Let  $S \subseteq \mathbb{Z}^3$  be the affine semigroup generated by

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$$

. Then we have  $X = \text{Spec}(\mathbb{C}[S])$ .

**Proposition 1.3.10**

Let  $k$  be an algebraically closed field. Let  $S$  be an affine semi-group with generators given by  $m_1, \dots, m_s$ . Then there is a bijection

$$\{p \in \text{Spec}(k[S]) \mid p \text{ is a closed point}\} \xrightarrow{1:1} \{\phi : S \rightarrow (\mathbb{C}, \cdot) \mid \phi \text{ is a monoid homomorphism}\}$$

given as follows.

- Each closed point  $p \in \text{Spec}(k[S])$  corresponds to the monoid homomorphism  $\gamma_p : S \rightarrow \mathbb{C}$  defined by

$$\gamma_p \left( \sum_{i=1}^s \lambda_i m_i \right) = \sum_{i=1}^s \lambda_i x_i(p)$$

where  $x_i \in \mathbb{C}[S]$  is the  $i$ th coordinate function.

- Each monoid homomorphism  $\gamma : S \rightarrow (\mathbb{C}, \cdot)$  corresponds to the point  $(\gamma(m_1), \dots, \gamma(m_s))$ .

**Proposition 1.3.11**

Let  $S_1, S_2$  be affine semigroups. Let  $\phi : \text{Spec}(\mathbb{C}[S_1]) \rightarrow \text{Spec}(\mathbb{C}[S_2])$  be a morphism. Then  $\phi$  is toric if and only if it is induced by a semigroup homomorphism  $S_1 \rightarrow S_2$ .

**1.4 Conditions for Normality in Affine Toric Varieties****Definition 1.4.1 (Saturation of an Affine Semigroup)**

Let  $S$  be an affine semigroup. Define the saturation of  $S$  to be

$$S_{\text{sat}} = (\mathbb{R}_{\geq 0} S) \cap G(S)$$

**Proposition 1.4.2**

Let  $S$  be an affine semigroup. Then we have

$$\overline{k[S]} = k[S_{\text{sat}}]$$

**Definition 1.4.3 (Saturated Affine Semigroup)**

Let  $S$  be an affine semigroup. We say that  $S$  is saturated if  $S = S_{\text{sat}}$ .

**Lemma 1.4.4**

Let  $\sigma \subseteq \mathbb{R}^n$  be a strongly convex rational polyhedral cone. Then  $\sigma \cap S$  is saturated.

**Lemma 1.4.5**

Let  $X = \text{Spec}(\mathbb{C}[S])$  be an affine toric variety over  $\mathbb{C}$  where  $S$  is an affine semigroup. Then the normalization of  $X$  is given by

$$\tilde{X} = \text{Spec}(\mathbb{C}[S_{\text{sat}}])$$

together with the normalization map given by the inclusion  $\mathbb{C}[S] \hookrightarrow \mathbb{C}[S_{\text{sat}}]$ .

**1.5 Equivalent Characterizations of Projective Toric Varieties****Definition 1.5.1 (Projective Toric Varieties Associated to Characters)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Define the projective toric variety associated to  $\mathcal{A}$  by

$$X_{\mathcal{A}} = \overline{\text{im}(\pi \circ \Phi_{\mathcal{A}})}$$

where  $\Phi_{\mathcal{A}} : T \rightarrow \mathbb{A}_{\mathbb{C}}^s$  is the map defined by  $\Phi_{\mathcal{A}}(t) = (\chi_1(t), \dots, \chi_s(t))$  and  $\pi : \mathbb{A}_{\mathbb{C}}^s \rightarrow \mathbb{CP}^{s-1}$  is the projection map.

**Proposition 1.5.2**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Then the following are equivalent.

- $Y_{\mathcal{A}}$  is the affine cone of  $X_{\mathcal{A}}$ .
- $\mathbb{I}(Y_{\mathcal{A}}) = \mathbb{I}^H(X_{\mathcal{A}})$ .
- $\mathbb{I}(Y_{\mathcal{A}})$  is a homogeneous ideal.
- There exists  $u \in N$  and  $k \in \mathbb{N}^+$  such that  $\langle \chi_i, u \rangle = k$  for  $1 \leq i \leq s$ .

**Proposition 1.5.3**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Then the following are true.

- We have

$$\chi(T_{X_{\mathcal{A}}}) = \mathbb{Z}\mathcal{A} \cap \left\{ (a_1, \dots, a_s) \in \mathbb{Z}^s \mid \sum_{i=1}^s a_i = 0 \right\}$$

- The dimension of the torus is given by

$$\dim(T_{X_{\mathcal{A}}}) = \begin{cases} \text{rank}(\mathbb{Z}\mathcal{A}) - 1 & \text{if } \mathbb{I}(Y_{\mathcal{A}}) \text{ is a homogeneous ideal} \\ \text{rank}(\mathbb{Z}\mathcal{A}) & \text{otherwise} \end{cases}$$

**Proposition 1.5.4**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Then we have

$$X_{\mathcal{A}} \cap U_i = Y_{\mathcal{A} \setminus \{\chi_i\}}$$

**Proposition 1.5.5**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$  be characters of  $T$ . Let  $J = \{j \in \{1, \dots, s\} \mid \chi_j \text{ is a vertex of } \text{Cone}(\text{Conv}(\mathcal{A}))\}$ . Then we have

$$X_{\mathcal{A}} = \bigcup_{j \in J} X_{\mathcal{A}} \cap U_j$$



## 2 Toric Varieties and Convex Geometry

### 2.1 Cones and Affine Toric Varieties

**Lemma 2.1.1** (Gordon's Lemma)

Let  $\sigma \subseteq \mathbb{R}^n$  be a rational polyhedral cone. Let  $A \subseteq \mathbb{Z}^n$  be a subgroup. Then  $\sigma \cap A$  is an affine semi-group.

**Definition 2.1.2** (Affine Semigroup Associated to a Cone)

Let  $\sigma \subseteq \mathbb{R}^n$  be a rational polyhedral cone. Define the associated affine semigroup to be

$$S_\sigma = \sigma^\vee \cap (\mathbb{R}^n)^*$$

**Definition 2.1.3** (Affine Toric Variety Associated to a Cone)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma \in \lambda(T) \otimes_{\mathbb{Z}} \mathbb{C}^\times$  be a rational polyhedral cone. Define the associated affine toric variety by

$$U_\sigma = \text{Spec}(\mathbb{C}[S_\sigma])$$

**Example 2.1.4**

The following are true.

- There is an isomorphism

$$\text{Spec}(\mathbb{C}[\{0\}^\vee \cap \mathbb{Z}^n]) \cong \text{Spec}(\mathbb{G}_m^n)$$

- Let  $e_1, e_2, e_3 \in \mathbb{R}^3$  be the standard basis vectors. Let  $\sigma = \text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3) \subseteq \mathbb{R}^3$ . Then we have

$$\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$$

- Let  $e_1, e_2, e_3 \in \mathbb{R}^3$  be the standard basis vectors. Let  $\sigma = \text{Cone}(e_2, e_3, 2e_1 + e_2 + e_3) \subseteq \mathbb{R}^3$ . Then we have

$$\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x_1, \dots, x_6]}{(x_1x_4 - x_2x_3, x_1x_5 - x_2^2, x_1x_6 - x_3^2)}$$

- Let  $e_1, e_2 \in \mathbb{R}^2$  be the standard basis vectors. Let  $\sigma = \text{Cone}(de_1 - e_2, e_2)$ . Then we have

$$U_\sigma = \widehat{C}_d$$

**Proof**

- The dual cone of  $\{0\}$  is  $\mathbb{R}^n$  and intersecting with  $\mathbb{Z}^n$  just gives itself. So  $\mathbb{C}[\mathbb{Z}^n] = \mathbb{C}[x_1, \dots, x_n, x_1^{-1}, \dots, x_n^{-1}]$  is just the coordinate ring of the standard  $n$ -torus.
- We compute the dual cone to be

$$\sigma^\vee = H_{(1,0,0)}^+ \cap H_{(0,1,0)}^+ \cap H_{(1,0,1)}^+ \cap H_{(0,1,1)}^+$$

A generating set of  $S_\sigma$  as a monoid is given by  $\{e_1, e_2, e_3, e_1 + e_2 - e_3\}$ . The kernel of the map  $\mathbb{Z}^4 \rightarrow \mathbb{Z}^3$  defined by  $e_i \mapsto e_i$  for  $1 \leq i \leq 3$  and  $e_4 \mapsto (1, 1, -1)$  is given by  $\mathbb{Z}\langle e_1 + e_2 - e_3 - e_4 \rangle$ . Thus we are done.

- We compute the dual cone to be

$$\sigma^\vee = H_{(2,1,1)}^+ \cap H_{(0,1,0)}^+ \cap H_{(0,0,1)}^+$$

A generating set of  $S_\sigma$  as a monoid is then given by  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (-1, 1, 1), (-1, 2, 0), (-1, 0, 2)\}$ . The kernel of the map  $\mathbb{Z}^6 \rightarrow \mathbb{Z}^3$  defined by  $e_i \mapsto e_i$  for  $1 \leq i \leq 3$  and  $e_4 \mapsto (-1, 1, 1), e_5 \mapsto (-1, 2, 0), e_6 \mapsto (-1, 0, 2)$  is given by  $\mathbb{Z}\langle e_1 - e_2 - e_3 + e_4, e_1 - 2e_2 + e_5, e_1 - 2e_3 + e_6 \rangle$ . The ideal associated to this lattice is given by  $(x_1x_4 - x_2x_3, x_1x_5 - x_2^2, x_1x_6 - x_3^2)$ . Thus we are done.

- The dual cone is given by

$$\sigma^\vee = H_{(d,-1)}^+ \cap H_{(0,1)}^+$$

A generating set for the monoid  $S_\sigma$  is given by  $\{e_1 + ie_2 \mid 0 \leq i \leq d\}$ . The kernel of the map  $\mathbb{Z}^{d+1} \rightarrow \mathbb{Z}^2$  defined by  $e_i \mapsto e_1 + ie_2$  is given by  $\mathbb{Z}\langle e_i + e_{j+1} - e_{i+1}e_j \mid 0 \leq i < j \leq d-1 \rangle$ . Notice that the lattice ideal of the kernel is precisely the ideal defining the rational normal cone. Hence we are done. ■

### Proposition 2.1.5

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma \in \lambda(T) \otimes_{\mathbb{Z}} \mathbb{C}^\times$  be a rational polyhedral cone. Then the following are equivalent.

- $\sigma$  is strongly convex.
- $T_{U_\sigma} = \lambda(T) \otimes_{\mathbb{Z}} \mathbb{C}^\times$ .
- $\dim(U_\sigma) = \text{rank}(\lambda(T)) = \dim(T)$ .

### Proposition 2.1.6

Let  $V$  be an affine toric variety. Then the following are equivalent.

- $V$  is normal.
- There exists a saturated affine semigroup such that  $V = \text{Spec}(\mathbb{C}[S])$ .
- There exists a strongly convex rational polyhedral cone  $\sigma$  such that  $V = \text{Spec}(\mathbb{C}[S_\sigma])$ .

### Example 2.1.7

The following are affine normal toric varieties.

- $\mathbb{V}(xy - zw) \subseteq \mathbb{A}_{\mathbb{C}}^4$ .
- For  $d \geq 1$ , the rational normal cone  $\widehat{C}_d$ .

### Proposition 2.1.8

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma \in \lambda(T)$  be a strongly convex rational polyhedral cone. Then  $U_\sigma$  has a fixed point if and only if  $\dim(\sigma) = \dim(T)$ . In this case, the maximal ideal corresponding to the fixed point is given by

$$\langle \chi \mid \chi \in S_\sigma \setminus \{0\} \rangle \subseteq \mathbb{C}[S_\sigma]$$

### Proposition 2.1.9

Let  $T_1, T_2$  be two tori. Let  $\sigma_1 \in T_1$  and  $\sigma_2 \in T_2$  be two strongly convex rational polyhedral cones. Let  $\phi : U_{\sigma_1} \rightarrow U_{\sigma_2}$  be a morphism. Then  $\phi$  is a toric morphism if and only if there exists a map  $\psi : \lambda(T_1) \rightarrow \lambda(T_2)$  such that the following are true.

- $\phi$  is induced by  $\psi \otimes_{\mathbb{Z}} \text{id}$ .
- $\psi(\sigma_1) \subseteq \sigma_2$ .

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma \in \lambda(T) \otimes_{\mathbb{Z}} \mathbb{C}^\times$  be a strongly convex rational polyhedral cone. Recall that  $\sigma$  is regular if it can be generated by a subset of a basis of  $\lambda(T)$ .

### Proposition 2.1.10

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma \in \lambda(T) \otimes_{\mathbb{Z}} \mathbb{C}^\times$  be a strongly convex rational polyhedral cone. Then  $U_\sigma$  is smooth if and only if  $\sigma$  is regular.

### Proposition 2.1.11

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma$  be a strongly convex rational polyhedral cone in  $\lambda(T)$ . Let  $\tau \subseteq \sigma$  be a face. Let  $H_\phi$  be the supporting hyperplane for  $\tau$  where  $\phi \in \sigma^\vee \cap \lambda(T)$ . Then we have

$$\mathbb{C}[S_\tau] = \mathbb{C}[S_\sigma]_\phi$$

**Example 2.1.12**

Let  $e_1, e_2, e_3 \subseteq \mathbb{R}^3$  be the standard basis vectors. Let  $\sigma = \text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3) \subseteq \mathbb{R}^3$ . Let  $\tau = \text{Cone}(e_1, e_1 + e_3)$ . Then we have

$$\mathbb{C}[S_\tau] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - 1)}$$

**Proof**

We already computed that  $\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$ . Notice that the supporting hyperplane for  $\tau$  is given by  $H_{(0,1,0)}^+$  (cross product of  $e_1$  and  $e_1 + e_3$  gives the normal to the plane  $\mathbb{R}\langle e_1, e_1 + e_3 \rangle$ ). Under the isomorphism  $\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$ , the element  $(0, 1, 0) \in S_\sigma$  corresponds to  $y$ . Hence

$$\mathbb{C}[S_\tau] = \left( \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)} \right)_y \cong \mathbb{C}[y, z, w, y^{-1}] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - 1)}$$

On the other hand, we may compute the coordinate ring  $\mathbb{C}[S_\tau]$  using the lattice ideal. Notice that the dual of  $\tau$  is given by

$$\tau^\vee = H_{(1,0,0)}^+ \cap H_{(1,0,1)}^+$$

A generating set for the monoid  $S_\tau$  is given by  $\{e_2, -e_2, e_3, e_1 - e_3\}$ . Then the kernel of the map  $\mathbb{Z}^4 \rightarrow \mathbb{Z}^3$  defined by  $e_i \mapsto e_i$  for  $2 \leq i \leq 3$  and  $e_4 \mapsto -e_2, e_1 \mapsto e_1 - e_3$  is given by  $\mathbb{Z}\langle e_2 + e_4 \rangle$  and so the coordinate ring is given by

$$\mathbb{C}[S_\tau] = \frac{\mathbb{C}[x, y, z, w]}{(yw - 1)}$$

and the two computations give isomorphic  $\mathbb{C}$ -algebras. ■

**2.2 Polytopes and Projective Toric Varieties**

We begin with the very ample case. Let  $T$  be a torus over  $\mathbb{C}$ . Let  $P \subseteq \chi(T) \otimes_{\mathbb{Z}} \mathbb{C}^*$  be a full-dimensional very ample polytope with lattice  $M$ . Suppose that  $|P \cap M| = s$  is finite. Then  $X_{P \cap M}$  is a projective toric variety.

**Definition 2.2.1 (The Projective Toric Variety Associated to a Lattice Polytope)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $P \subseteq \chi(T) \otimes_{\mathbb{Z}} \mathbb{C}^*$  be a full-dimensional lattice polytope with lattice  $M$ . Define the toric variety associated to  $P$  by

$$X_P = X_{(kP) \cap M}$$

for any  $k$  such that  $kP$  is very ample.

Such a  $k$  exists by Ewald-Wessels theorem, which says that  $kP$  is normal for all  $k \geq \dim(P) - 1$ .

**Proposition 2.2.2**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $P \subseteq \chi(T) \otimes_{\mathbb{Z}} \mathbb{C}^*$  be a full-dimensional very ample polytope with lattice  $M$ . Suppose that  $P \cap M = \{\chi_1, \dots, \chi_s\}$  is finite. Then the following are true.

- We have

$$X_{P \cap M} \cap U_i = U_{\sigma_i}$$

where  $\sigma_i$  is the strongly convex rational cone dual to  $\text{Cone}(P \cap M \setminus \{\chi_i\})$ .

- The torus is given by

$$T_{X_{P \cap M}} = T$$

**Proposition 2.2.3**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $P \subseteq \chi(T) \otimes_{\mathbb{Z}} \mathbb{C}^*$  be a full-dimensional lattice polytope with lattice  $M$ . Then the following are true.

- $X_P$  is normal.
- $X_P$  is projectively normal under the embedding induced by  $kP$  if and only if  $kP$  is normal.

## 2.3 Fans and Normal Toric Varieties

### Definition 2.3.1 (Toric Variety Associated to a Fan)

Let  $T$  be a torus. Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Define the scheme  $X_\Sigma$  associated to  $\Sigma$  by the following gluing data.

- The schemes  $\text{Spec}(\mathbb{C}[S_\sigma])$  for each  $\sigma \in \Sigma$ .
- For each  $\sigma_1, \sigma_2 \in \Sigma$ , an open subscheme

$$\text{Spec}(\mathbb{C}[S_{\sigma_1 \cap \sigma_2}]) \subseteq \text{Spec}(\mathbb{C}[S_{\sigma_1}]) \quad \text{and} \quad \text{Spec}(\mathbb{C}[S_{\sigma_2}])$$

- The gluing maps are the identity maps.

We need to check that the above data is sufficient to define a scheme for gluing.

### Proposition 2.3.2

Let  $T$  be a torus. Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then  $X_\Sigma$  is a normal, separated toric variety with torus  $T$ .

#### Proof

Sketch proof:

Since each cone in  $\Sigma$  is strongly convex, we know that  $\{0\} \in \Sigma$  and it is the face of every cone. Hence we have inclusions  $\text{Spec}(\mathbb{C}[S_{\{0\}}]) \hookrightarrow \text{Spec}(\mathbb{C}[S_\sigma])$  for every  $\sigma \in \Sigma$ . Hence each of the torii is glued by the identity. Thus  $\text{Spec}(\mathbb{C}[S_{\{0\}}])$  is a torus contained in  $X_\Sigma$ . For each  $\sigma \in \Sigma$ , we know that  $\text{Spec}(\mathbb{C}[S_{\{0\}}])$  acts on  $\text{Spec}(\mathbb{C}[S_\sigma])$  for each  $\sigma \in \Sigma$ . Hence they glue to an action on  $X_\Sigma$ . (Check: the action is compatible with intersections and patches).

Since each  $\text{Spec}(\mathbb{C}[S_\sigma])$  is irreducible, we must have  $X_\Sigma$  is irreducible. Hence  $X_\Sigma$  is a toric variety.

Since normality is a local condition, we conclude that  $X_\Sigma$  is normal.

Now notice that  $X_\Sigma$  is separated if and only if for any cones  $\sigma_1, \sigma_2 \in \Sigma$ , the product map  $\Delta : \text{Spec}(\mathbb{C}[S_{\sigma_1 \cap \sigma_2}]) \rightarrow \text{Spec}(\mathbb{C}[S_{\sigma_1}]) \times \text{Spec}(\mathbb{C}[S_{\sigma_2}])$  is Zariski closed. Now the map  $\Delta$  is induced by the  $\mathbb{C}$ -algebra homomorphism  $\Delta^* : \mathbb{C}[S_{\sigma_1}] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma_2}] \rightarrow \mathbb{C}[S_{\sigma_1 \cap \sigma_2}]$  by  $\chi^{m_1} \otimes \chi^{m_2} \mapsto \chi^{m_1 + m_2}$ . By the separation lemma, we know that  $S_{\sigma_1} + S_{\sigma_2} = S_{\sigma_1 \cap \sigma_2}$ . Thus  $\Delta^*$  is surjective. Then we have

$$\mathbb{C}[\text{im}(\Delta)] = \mathbb{C}[S_{\sigma_1}] \otimes_{\mathbb{C}} \mathbb{C}[S_{\sigma_2}] / \ker(\Delta^*)$$

so that  $\text{im}(\Delta)$  is Zariski closed. ■

### Example 2.3.3 (Projective Space)

Let  $e_1, \dots, e_n \in \mathbb{R}^n$  be the standard basis vectors of  $\mathbb{R}^n$ . Denote  $e_{n+1} = -\sum_{i=1}^n e_i$ . Let

$$\Sigma = \{\text{Cone}(e_{i_1}, \dots, e_{i_k}) \mid 1 \leq k \leq n+1, 1 < i_1 < \dots < i_k \leq n+1\}$$

Then we have

$$X_\Sigma = \mathbb{P}_{\mathbb{C}}^n$$

### Example 2.3.4 (Hirzebruch Surface)

Let  $r \in \mathbb{N}$ . Let  $e_1, e_2 \in \mathbb{R}^2$  be the standard basis vectors of  $\mathbb{R}^2$ . Let

$$\Sigma = \{\text{Cone}(e_1, e_2), \text{Cone}(e_1, -e_2), \text{Cone}(-e_1 + re_2, -e_2), \text{Cone}(-e_1 + re_2, e_1)\}$$

Define the Hirzebruch surface by

$$\mathcal{H}_r = X_\Sigma$$

**Theorem 2.3.5 ((Sumihiro))**

Let  $X$  be a normal separated variety equipped with the action of a torus  $T$ . Let  $p \in X$ . Then there exists an affine open neighbourhood  $U$  of  $p$  such that  $U$  is  $T$ -invariant.

**Corollary 2.3.6**

Let  $X$  be a normal separated toric variety. Then there exists a rational fan  $\Sigma$  in  $\lambda(T_X)$  such that  $X = X_\Sigma$ .

Intuition: Adding a new cone to a fan partially compactifies.

**Proposition 2.3.7**

Let  $T_1, T_2$  be a tori over  $\mathbb{C}$ . Let  $\Sigma_1, \Sigma_2$  be rational fans in  $\lambda(T_1), \lambda(T_2)$  respectively. Then we have

$$X_{\Sigma_1 \times \Sigma_2} \cong X_{\Sigma_1} \times X_{\Sigma_2}$$

**Proposition 2.3.8**

Let  $T$  be a torus. Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then the following are true.

- $X_\Sigma$  is smooth if and only if  $\Sigma$  is smooth.
- $X_\Sigma$  is complete if and only if  $\Sigma$  is complete.

**Proposition 2.3.9**

Let  $T_1, T_2$  be a tori over  $\mathbb{C}$ . Let  $\Sigma_1, \Sigma_2$  be rational fans in  $\lambda(T_1), \lambda(T_2)$  respectively. Let  $\phi : X_{\Sigma_1} \rightarrow X_{\Sigma_2}$  be a morphism. Then  $\phi$  is a toric morphism if and only if there exists a  $\mathbb{Z}$ -linear map  $f : \lambda(T_1) \rightarrow \lambda(T_2)$  such that the following are true.

- For each cone  $\sigma \in \Sigma_1$ , there exists a cone  $\tau \in \Sigma_2$  such that  $f(\sigma) \subseteq \tau$ .
- $\phi|_{T_1} = f \otimes_{\mathbb{Z}} \text{id}_{\mathbb{C}^*}$ .

**Definition 2.3.10 (Ray Generators)**

Let  $\Sigma \subseteq \mathbb{R}^n$  be a rational fan. Let  $\rho \in \Sigma(1)$  be a ray. Denote  $u_\rho$  by the ray generator of  $\rho$ .

**Proposition 2.3.11**

Let  $T$  be a torus. Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then the following are equivalent.

- $X_\Sigma$  has no torus factors.
- $\Gamma(X_\Sigma, \mathcal{O}_{X_\Sigma})^* = \mathbb{C}^*$ .
- The set  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ .

## 2.4 The Orbit-Cone Correspondence

**Definition 2.4.1 (Distinguished Point of a Cone)**

Let  $\sigma \subseteq \mathbb{R}^n$  be a cone. Let  $\gamma : S_\sigma \rightarrow \mathbb{C}$  be the monoid homomorphism defined by

$$\gamma(m) = \begin{cases} 1 & \text{if } m \in S_\sigma \cap \sigma^\perp = \sigma^\perp \cap (\mathbb{R}^n)^* \\ 0 & \text{otherwise} \end{cases}$$

Define the distinguished point of  $\sigma$  to be the closed point in  $\text{Spec}(\mathbb{C}[S_\sigma])$  associated to  $\gamma$ . We denote it by  $p_\sigma$ .

**Lemma 2.4.2**

Let  $\Sigma$  be a rational fan in  $\mathbb{R}^n$ . Let  $\sigma \in \Sigma$  be a cone. Then we have

$$\dim(\text{Orb}_{T_{X_\Sigma}}(p_\sigma)) = n - \dim(\sigma)$$

**Theorem 2.4.3** (The Orbit-Cone Correspondence)

Let  $\Sigma$  be a rational fan in  $\mathbb{R}^n$ . Then there is a one to one correspondence

$$\Sigma \xleftrightarrow{1:1} X_\Sigma / T_{X_\Sigma}$$

given by  $\sigma \mapsto \text{Orb}_{T_{X_\Sigma}}(p_\sigma)$ .

**Proposition 2.4.4**

Let  $\Sigma$  be a rational fan in  $\mathbb{R}^n$ . Let  $\sigma \in \Sigma$  be a cone. Then we have

$$\text{Spec}(\mathbb{C}[S_\sigma]) = \bigcup_{\tau \text{ is a face of } \sigma} \text{Orb}_{T_{X_\Sigma}}(\tau)$$

### 3 The Cox Ring

#### 3.1 Basic Definitions

**Definition 3.1.1 (The Cox Ring)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Define the Cox ring of  $X_\Sigma$  to be the algebra

$$\text{Cox}(X_\Sigma) = \mathbb{C}[x_\rho \mid \rho \in \Sigma(1)]$$

together with the grading given by

$$\deg \left( \prod_{\rho \in \Sigma(1)} x_\rho^{a_\rho} \right) = \left[ \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \right] \in \text{Cl}(X_\Sigma)$$

#### 3.2 Toric Varieties as a Quotient of the Cox Ring

**Definition 3.2.1 (The Irrelevant Ideal)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Define the irrelevant ideal by

$$B(\Sigma) = \left\langle \prod_{\rho \notin \sigma(1)} x_\rho \mid \sigma \in \Sigma_{\max} \right\rangle \subseteq \text{Cox}(X_\Sigma)$$

**Definition 3.2.2 (The Exceptional Set)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Define the exceptional set to be

$$Z(\Sigma) = \mathbb{V}^H(B(\Sigma)) \subseteq X_\Sigma$$

#### 3.3 The Toric Nullstellensatz

The projective nullstellensatz can be viewed as a more general phenomenon in the toric setting.

**Theorem 3.3.1 (The Toric Weak Nullstellensatz)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Let  $I \subseteq \text{Cox}(X_\Sigma)$  be a homogeneous ideal. Then we have

$$\mathbb{V}^H(I) \subseteq Z(\Sigma) \quad \text{if and only if} \quad B(\Sigma) \subseteq \sqrt{I}$$

**Theorem 3.3.2 (The Toric Nullstellensatz)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Then there is an inclusion reversing bijection

$$\left\{ I \subseteq \text{Cox}(X_\Sigma) \mid \begin{array}{l} \text{\textit{I is a radical homogeneous}} \\ \text{\textit{such that } } I \subseteq B(\Sigma) \end{array} \right\} \xleftrightarrow{1:1} \left\{ \text{Closed Subsets of } X_\Sigma \right\}$$

given by  $I \mapsto \mathbb{V}^H(I)$  and  $Y \mapsto \mathbb{I}^H(Y)$ .

## 4 Divisors and Torus Invariant Divisors

### 4.1 Divisors on Toric Varieties

#### Definition 4.1.1 (The Divisor of a Ray)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $\rho \in \Sigma(1)$  be a ray. Define the associated divisor of  $\rho$  by

$$D(\rho) = \overline{\text{Orb}_T(p_\rho)} \in \text{Div}(X_\Sigma)$$

#### Proposition 4.1.2

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $\chi$  be a character of  $T$ . Then the following are true.

- $\chi$  is a rational function on  $X_\Sigma$ .
- The divisor of  $\chi$  is given by

$$\text{div}(\chi) = \sum_{\rho \in \Sigma(1)} \langle \chi, u_\rho \rangle D(\rho)$$

#### Definition 4.1.3 ( $T$ -Invariant Divisor Group)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Define the  $T$ -invariant divisor group of  $X_\Sigma$  to be the subgroup

$$\text{Div}_T(X_\Sigma) = \bigoplus_{\rho \in \Sigma(1)} \mathbb{Z} \cdot D(\rho) \leq \text{Div}(X_\Sigma)$$

#### Proposition 4.1.4

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then the following are true.

- There is an exact sequence of the form

$$\chi(T) \longrightarrow \text{Div}_T(X_\Sigma) \longrightarrow \text{Cl}(X_\Sigma) \longrightarrow 0$$

where the first map is given by  $\chi \mapsto \text{div}(\chi)$ .

- If  $X_\Sigma$  has no torus factors, then the above sequence is exact on the left:

$$0 \longrightarrow \chi(T) \longrightarrow \text{Div}_T(X_\Sigma) \longrightarrow \text{Cl}(X_\Sigma) \longrightarrow 0$$

#### Example 4.1.5

Let  $d \geq 1$ . Then we have

$$\text{Cl}(\widehat{C}_d) = \mathbb{Z}$$

#### Proof

Recall that the affine toric variety is the toric variety associated to the cone  $\text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3)$ . It has four 1-dimensional faces given by  $e_1, e_2, e_1 + e_3, e_2 + e_3$ . Now the character lattice for the torus is  $\mathbb{Z}^3$  with generators  $e_1, e_2, e_3$ . Then we compute that  $\text{div}(e_1) = \overline{O(e_1)} + \overline{O(e_1 + e_3)}$  and

etc. By the above proposition, it suffices to compute the cokernel of the matrix  $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ . So

we are done. ■

#### Example 4.1.6

We have

$$\text{Cl}\left(\text{Spec}\left(\frac{\mathbb{C}[x_0, \dots, x_d]}{(x_i x_{j+1} - x_j x_{i+1} \mid 0 \leq i < j \leq d-1)}\right)\right) = \mathbb{Z}/d\mathbb{Z}$$



**Proof**

Recall that the rational normal cone is given by the cone  $\sigma = \text{Cone}(e_2, de_1 - e_2)$ . It has two 1-dimensional faces whose generators are given by  $e_2$  and  $de_1 - e_2$ . Now the character lattice of  $S_\sigma$  is  $\mathbb{Z}^2$  with generators  $e_1$  and  $e_2$ . Then we compute that

$$\text{div}(e_1) = \langle e_1, e_2 \rangle \overline{O(e_2)} + \langle e_1, de_1 - e_2 \rangle \overline{O(de_1 - e_2)} = d\overline{O(de_1 - e_2)}$$

and similarly that  $\text{div}(e_2) = \overline{O(e_2)} - \overline{O(de_1 - e_2)}$ . By the above proposition, it suffices to compute the cokernel of  $\begin{pmatrix} 0 & 1 \\ d & -1 \end{pmatrix}$ . Hence we are done. ■

**Example 4.1.7**

Let  $r \in \mathbb{N}$ . Then we have

$$\text{Cl}(\mathcal{H}_r) \cong \mathbb{Z}^2$$

**Proof**

Recall that the cone of  $\mathcal{H}_r$  is given by

$$\Sigma = \{\text{Cone}(e_1, e_2), \text{Cone}(e_1, -e_2), \text{Cone}(-e_1 + re_2, -e_2), \text{Cone}(-e_1 + re_2, e_1)\}$$

Clearly the rays  $e_1, e_2, -e_2, -e_1 + re_2$  span  $\lambda(T)$ . The inclusion map  $\chi(T) \rightarrow \text{Div}_T(\mathcal{H}_r)$  is as follows. Let  $\chi_1(s, t) = s$  and  $\chi_2(s, t) = t$  be the basis of  $\chi(T)$ . Then we have

$$\text{div}(\chi_1) = D(e_1) - D(-e_1 + re_2) \quad \text{and} \quad \text{div}(\chi_2) = D(e_2) - D(-e_2) + rD(-e_1 + re_2)$$

The matrix representing the  $\mathbb{Z}$ -linear map and its Smith normal form is given by

$$\text{SNF} \left( \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -1 \\ -1 & r \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

The cokernel is precisely the class group and so we have

$$\text{Cl}(\mathcal{H}_r) \cong \mathbb{Z} \cdot [D(e_1)] \oplus \mathbb{Z} \cdot [D(e_2)]$$
■

**Proposition 4.1.8**

Let  $T_1, T_2$  be tori over  $\mathbb{C}$ . Let  $\Sigma_1, \Sigma_2$  be rational fans over  $\lambda(T_1)$  and  $\lambda(T_2)$  respectively. Then we have

$$\text{Cl}(X_{\Sigma_1 \times \Sigma_2}) \cong \text{Cl}(X_{\Sigma_1}) \oplus \text{Cl}(X_{\Sigma_2})$$

**Definition 4.1.9** (*T-Invariant Cartier Divisor Group*)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Define the  $T$ -invariant Cartier divisor group of  $X_\Sigma$  to be the subgroup

$$\text{CDiv}_T(X_\Sigma) = \left\{ D \in \text{CDiv}(X_\Sigma) \mid D \sim \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{Div}(X_\Sigma) \right\}$$

**Proposition 4.1.10**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then the following are true.

- There is an exact sequence of the form

$$\chi(T) \longrightarrow \text{CDiv}_T(X_\Sigma) \longrightarrow \text{CCl}(X_\Sigma) \longrightarrow 0$$

where the first map is given by  $\chi \mapsto \text{div}(\chi)$  and the second map is the restriction of the

quotient map from the group of divisors to the class group.

- If  $X_\Sigma$  has no torus factors, then the above sequence is exact on the left:

$$0 \longrightarrow \chi(T) \longrightarrow \text{CDiv}_T(X_\Sigma) \longrightarrow \text{CCl}(X_\Sigma) \longrightarrow 0$$

#### Proposition 4.1.11

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma$  be a strongly convex rational cone in  $\lambda(T)$ . Then we have

$$\text{Cl}(U_\sigma) = 0$$

#### Example 4.1.12

The following are true.

- $\text{CCl}(\mathbb{V}(xy - zw)) = 0$ .
- For  $d \geq 1$ , we have  $\text{CCl}(\widehat{C}_d) = 0$

#### Proposition 4.1.13

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $\sigma \in \Sigma$  be a cone such that  $\dim(\sigma) = \dim(T)$ . Then  $\text{CCl}(X_\Sigma)$  is a free abelian group.

#### Proposition 4.1.14

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then the following are equivalent.

- $X_\Sigma$  is smooth.
- $\text{Div}(X_\Sigma) = \text{CDiv}(X_\Sigma)$
- $\text{Cl}(X_\Sigma) \cong \text{CCl}(X_\Sigma)$ .

#### Example 4.1.15

The following are true.

- $\text{CCl}(\mathbb{P}_{\mathbb{C}}^n) \cong \mathbb{Z}$ .
- For  $r \in \mathbb{N}$ , we have  $\text{CCl}(\mathcal{H}_r) \cong \mathbb{Z}^2$ .

#### Proposition 4.1.16

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Then the following are equivalent.

- $\Sigma$  is simplicial.
- $X_\Sigma$  is an orbifold.
- $\text{CCl}(X_\Sigma)$  has finite index as a subgroup of  $\text{Cl}(X_\Sigma)$ .

#### Proposition 4.1.17

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{Div}_T(X_\Sigma)$  be a Weil divisor. Then the following are equivalent.

- $D$  is a Cartier divisor.
- $D|_{U_\sigma} \in \text{Div}(U_\sigma)$  is principal for all  $\sigma \in \Sigma$ .
- For each  $\sigma \in \Sigma$ , there exists  $\chi_\sigma \in \chi(T)$  such that  $\langle \chi_\sigma, u_\rho \rangle = -a_\rho$  for all  $\rho \in \sigma(1)$ .
- For each  $\sigma \in \Sigma_{\max}$ , there exists  $\chi_\sigma \in \chi(T)$  such that  $\langle \chi_\sigma, u_\rho \rangle = -a_\rho$  for all  $\rho \in \sigma(1)$ .

#### Definition 4.1.18 (Cartier Data of Toric Variety)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. A Cartier data for  $D$  is a collection  $\{m_\sigma\}_{\sigma \in \Sigma}$  such that

$$\langle \chi_\sigma, u_\rho \rangle = -a_\rho$$

for all  $\rho \in \sigma(1)$ .

Colimit description of CCl.

**Definition 4.1.19** (The Polyhedron Associated to a Divisor)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{Div}_T(X_\Sigma)$ . Define the associated polyhedron by

$$P_D = \{\chi \in \chi(T) \otimes_{\mathbb{Z}} \mathbb{R} \mid \langle \chi, u_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Sigma(1)\}$$

**Lemma 4.1.20**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D \in \text{Div}_T(X_\Sigma)$ . If  $X_\Sigma$  is complete, then  $P_D$  is a convex polytope.

**Proposition 4.1.21**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D \in \text{Div}_T(X_\Sigma)$ . Then we have

$$\Gamma(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) \cong \bigoplus_{\chi \in \chi(T), \text{div}(\chi) + D \geq 0} \mathbb{C} \cdot \chi = \bigoplus_{\chi \in P_D \cap \chi(T)} \mathbb{C} \cdot \chi$$

## 4.2 Support Functions

**Definition 4.2.1** (Support Functions)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $\varphi : |\Sigma| \rightarrow \mathbb{R}$  be a function. We say that  $\varphi$  is a support function if for each  $\sigma \in \Sigma$ , we have that  $\varphi|_\sigma : \sigma \rightarrow \mathbb{R}$  is a linear map.

**Definition 4.2.2** (Integral Support Functions)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $\varphi : |\Sigma| \rightarrow \mathbb{R}$  be a support function. We say that  $\varphi$  is integral if

$$\varphi(|\Sigma| \cap \lambda(T)) \subseteq \mathbb{Z}$$

Denote the set of all integral support functions by

$$\text{SF}(\Sigma, \lambda(T)) = \{\varphi : |\Sigma| \rightarrow \mathbb{R} \mid \varphi \text{ is an integral support function}\}$$

**Definition 4.2.3** (The Associated Support Function)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{Div}_T(X_\Sigma)$ . Let  $\{m_\sigma\}_{\sigma \in \Sigma}$  be a Cartier data of  $D$ . Define the associated support function by

$$\varphi_D(u) = \langle m_\sigma, u \rangle \quad \text{for } u \in \sigma$$

**Proposition 4.2.4**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{Div}_T(X_\Sigma)$ . Then there is a group isomorphism

$$\text{CDiv}_T(X_\Sigma) \cong \text{SF}(\Sigma, \lambda(T))$$

given by  $D \mapsto \varphi_D$ .

## 4.3 Basepoint Free and Ample Divisors

**Proposition 4.3.1**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$ . Suppose that  $\Sigma_{\max} = \Sigma(\dim(T))$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{CDiv}_T(X_\Sigma)$ . For each  $\sigma \in \Sigma$ , let  $\chi_\sigma \in \chi(T)$  be the unique character such that

$$\langle \chi_\sigma, u_\rho \rangle = -a_\rho$$

for all  $\rho \in \Sigma(1)$ . Then  $D$  is basepoint free if and only if  $m_\sigma \in P_D$  for all  $\sigma \in \Sigma(\dim(T))$ .

Recall that if  $|\Sigma|$  is a cone that is convex and full-dimensional, then we have

$$|\Sigma| = \text{Cone}(\{u_\rho \mid \rho \in \Sigma(1)\}) = \bigcup_{\sigma \in \Sigma(n)} \sigma$$

In particular, we have  $\Sigma_{\max} = \Sigma(n)$ . Also recall that a function  $f$  on a convex set is concave if we have

$$f(tu + (1-t)v) \geq tf(u) + (1-t)f(v)$$

for all  $u, v$  in the domain and  $t \in [0, 1]$ .

#### Lemma 4.3.2

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  with convex support and is full-dimensional. Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Let  $\{m_\sigma\}_{\sigma \in \Sigma}$  be a Cartier data of  $D$ . Then the following are equivalent.

- $\varphi_D$  is concave.
- $\varphi_D(u) \leq \langle m_\sigma, u \rangle$  for all  $u \in |\Sigma|$  and  $\sigma \in \Sigma(\dim(T))$ .
- $\varphi_D(u) = \min\{\langle m_\sigma, u \rangle \mid \sigma \in \Sigma(\dim(T))\}$  for all  $u \in |\Sigma|$ .

#### Lemma 4.3.3

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  with convex support and is full-dimensional. Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Then we have

$$P_D = \{\chi \in \chi(T) \mid \varphi_D(u) \leq \langle m, u \rangle \text{ for all } u \in |\Sigma|\}$$

#### Proposition 4.3.4

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  with convex support and is full-dimensional. Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Let  $\{m_\sigma\}_{\sigma \in \Sigma}$  be a Cartier data of  $D$ . Then the following are equivalent.

- $D$  is basepoint free.
- $m_\sigma \in P_D$  for all  $\sigma \in \Sigma(\dim(T))$ .
- $\varphi_D$  is concave.

#### Proposition 4.3.5

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete rational fan in  $\lambda(T)$ . Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Let  $\{m_\sigma\}_{\sigma \in \Sigma}$  be a Cartier data of  $D$ . Then the following are equivalent.

- $D$  is basepoint free.
- $P_D = \text{Conv}(\{m_\sigma \mid \sigma \in \Sigma(\dim(T))\})$ .
- $V(P_D) = \{m_\sigma \mid \sigma \in \Sigma(\dim(T))\}$ .
- $\varphi_D(u) = \min\{\langle m, u \rangle \mid m \in P_D\}$  for all  $u \in \lambda(T)$ .

#### Definition 4.3.6 (Strictly Concave Associated Support Functions)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete rational fan in  $\lambda(T)$ . Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Let  $\{m_\sigma\}_{\sigma \in \Sigma}$  be a Cartier data of  $D$ . We say that  $\varphi_D$  is strictly concave if the following are true.

- $\varphi_D$  is concave.
- For all  $\sigma \in \Sigma(\dim(T))$ , we have

$$\langle m_\sigma, u \rangle = \varphi_D(u)$$

if and only if  $u \in \sigma$ .

#### Lemma 4.3.7

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete rational fan in  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho) \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Let  $\{m_\sigma\}_{\sigma \in \Sigma}$  be a Cartier data of  $D$ . Then the following are equivalent.

- $\varphi_D$  is strictly concave.
- $\varphi_D(u) < \langle m_\sigma, u \rangle$  for all  $u \in |\Sigma| \setminus \sigma$  and  $\sigma \in \Sigma(\dim(T))$ .
- $\varphi_D$  is concave and  $m_\sigma \neq m_\tau$  when  $\sigma \neq \tau \in \Sigma(\dim(T))$ .

- $\langle m_\sigma, u_\rho \rangle > -a_\rho$  for all  $\rho \in \Sigma(1) \setminus \sigma(1)$  and  $\sigma \in \Sigma(\dim(T))$ .
- $\varphi_D(u+v) > \varphi_D(u) + \varphi_D(v)$  for all  $u, v \in |\Sigma|$  and  $u, v$  does not lie in the same cone.

**Proposition 4.3.8**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete rational fan in  $\lambda(T)$ . Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Then  $D$  is ample if and only if  $\varphi_D$  is strictly convex.

**Proposition 4.3.9**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete rational fan in  $\lambda(T)$ . Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. If  $\dim(T) \geq 2$  and  $D$  is ample, then  $kD$  is very ample for all  $k \geq \dim(T) - 1$ .

**Proposition 4.3.10**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a smooth complete rational fan in  $\lambda(T)$ . Let  $D \in \text{CDiv}_T(X_\Sigma)$  be a Cartier divisor. Then  $D$  is ample if and only if it is very ample.

**Theorem 4.3.11 (Toric Chow Lemma)**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete rational fan in  $\lambda(T)$ . Then there exists a refinement  $\Sigma'$  of  $\Sigma$  such that  $X_{\Sigma'}$  is projective and the induced toric morphism  $X_{\Sigma'} \rightarrow X_\Sigma$  is proper and birational.

**4.4 Divisors of Toric Varieties of a Polytope****Proposition 4.4.1**

Let  $T$  be a torus over  $\mathbb{C}$ . There is a one to one correspondence

$$\left\{ P \subseteq \chi(T) \otimes_{\mathbb{Z}} \mathbb{R} \mid \begin{array}{c} P \text{ is a full-} \\ \text{dimensional laticc polytope} \end{array} \right\} \xleftrightarrow{1:1} \left\{ (X_\Sigma, D) \mid \begin{array}{c} \Sigma \text{ is a complete fan in } \lambda(T) \\ \text{and } D \in \text{CDiv}_T(X_\Sigma) \text{ is ample} \end{array} \right\}$$

given by  $P \mapsto (X_P, D_P)$  and  $(X_\Sigma, D) \mapsto P_D$ .

## 5 Sheaves and Bundles on Normal Toric Varieties

### 5.1 Sheaves on the Cox Ring

Recall that every quasi-coherent sheaves of modules over  $\mathbb{P}^n$  is given by the associated sheaf of a graded module over the polynomial ring. There is also a toric generalization of this result.

**Proposition 5.1.1**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Then the following are true.

- The functor

$$\widetilde{(-)} : \mathbf{GrMod}_{\mathrm{Cox}(X_\Sigma)} \rightarrow \mathbf{QCoh}_{X_\Sigma}$$

is essentially surjective.

- The functor

$$\widetilde{(-)} : \mathbf{FGMod}_{\mathrm{Cox}(X_\Sigma)} \rightarrow \mathbf{Coh}_{X_\Sigma}$$

is essentially surjective.

**Proposition 5.1.2**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a rational fan in  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Let  $D \in \mathrm{Div}_T(X_\Sigma)$ . Then there is an isomorphism of  $\mathcal{O}_X$ -modules

$$\mathcal{O}_X(D) \cong \mathcal{O}_X([D]) = \widetilde{S_{[D]}}$$

### 5.2 Toric Vector Bundles

## 6 Projective Morphisms and Projective Bundles

## 7 The Sheaf of Differential Forms

### 7.1

#### Proposition 7.1.1

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a regular fan on  $\lambda(T)$ . Then there is an exact sequence

$$0 \longrightarrow \Omega_{X_\Sigma}^1 \xrightarrow{\alpha} M \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma} \xrightarrow{\beta} \bigoplus_{\rho \in \Sigma(1)} i_* \mathcal{O}_{D(\rho)} \longrightarrow 0$$

where the maps are given as follows.

- $\alpha$  is defined by gluing the maps  $\Omega_{\mathbb{C}[S_\sigma]/\mathbb{C}}^1 \rightarrow M \otimes_{\mathbb{Z}} \mathbb{C}[S_\sigma]$  given by  $d\chi \mapsto m \otimes \chi$ .
- $\beta$  is the direct sum of the maps induced by the inclusion  $i : D(\rho) \hookrightarrow X_\Sigma$ .

#### Proposition 7.1.2

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then there is an exact sequence

$$0 \longrightarrow \widehat{\Omega}_{X_\Sigma}^1 \xrightarrow{\alpha \vee \vee} M \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma} \xrightarrow{\beta} \bigoplus_{\rho \in \Sigma(1)} i_* \mathcal{O}_{D(\rho)}$$

#### Proposition 7.1.3

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . If  $\Sigma$  is simplicial, then there is a short exact sequence

$$0 \longrightarrow \widehat{\Omega}_{X_\Sigma}^1 \xrightarrow{\alpha \vee \vee} M \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma} \xrightarrow{\beta} \bigoplus_{\rho \in \Sigma(1)} i_* \mathcal{O}_{D(\rho)} \longrightarrow 0$$

#### Proposition 7.1.4

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Then there is a natural isomorphism

$$\widehat{\Omega}_{X_\Sigma}^1 \cong \widehat{\Omega_{\text{Cox}(X_\Sigma)}^1}$$

#### Proposition 7.1.5

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then there is an exact sequence

$$0 \longrightarrow \widehat{\Omega}_{X_\Sigma}^p \xrightarrow{\alpha = \bigwedge_{i=1}^p} \bigwedge_{i=1}^p M \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma} \xrightarrow{\beta_p} \bigoplus_{\rho \in \Sigma(1)} \bigwedge_{i=1}^{p-1} (\rho^\perp \cap \chi(T)) \otimes_{\mathbb{Z}} i_* \mathcal{O}_{D(\rho)}$$

#### Proposition 7.1.6

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\sigma$  be a strongly convex rational polyhedral cone over  $\lambda(T)$ . Then we have an isomorphism

$$\Gamma(U_\sigma, \widehat{\Omega}_{U_\sigma}^p) \cong \bigoplus_{\chi \in \sigma^\vee \cap \chi(T)} \bigwedge_{i=1}^p V_\sigma(\chi) \cdot \chi$$

where we have

$$V_\sigma(\chi) = \mathbb{C} \left\langle \chi_0 \in \chi(T) \mid \chi_0 \text{ lies in the minimal face of } \sigma \text{ containing } \chi \right\rangle$$

### 7.2 The Toric Euler Exact Sequence

#### Theorem 7.2.1 (The Toric Euler Exact Sequence)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a simplicial fan on  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Then there is an exact sequence

$$0 \longrightarrow \widehat{\Omega}_{X_\Sigma}^1 \longrightarrow \bigoplus_{\rho \in \Sigma(1)} \mathcal{O}_{X_\Sigma}(-D(\rho)) \longrightarrow \text{Cl}(X_\Sigma) \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma} \longrightarrow 0$$



**Corollary 7.2.2**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a regular fan on  $\lambda(T)$  such that  $\{u_\rho \mid \rho \in \Sigma(1)\}$  spans  $\lambda(T)$ . Then there are exact sequences

$$0 \longrightarrow \Omega_{X_\Sigma}^1 \longrightarrow \bigoplus_{\rho \in \Sigma(1)} \mathcal{O}_{X_\Sigma}(-D(\rho)) \longrightarrow \text{Cl}(X_\Sigma) \otimes_{\mathbb{Z}} \mathcal{O}_{X_\Sigma} \longrightarrow 0$$

$$0 \longrightarrow \mathcal{O}_{X_\Sigma} \longrightarrow \bigoplus_{\rho \in \Sigma(1)} \mathcal{O}_{X_\Sigma}(D(\rho)) \longrightarrow \mathcal{T}_{X_\Sigma} \longrightarrow 0$$

**7.3 The Canonical Bundle****Proposition 7.3.1**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then there is a natural isomorphism

$$\mathcal{K}_{X_\Sigma} \cong \mathcal{O}_{X_\Sigma} \left( - \sum_{\rho \in \Sigma(1)} D(\rho) \right)$$

**Corollary 7.3.2**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then the canonical divisor of  $X_\Sigma$  is  $T$ -invariant and is given by

$$K_{X_\Sigma} = - \sum_{\rho \in \Sigma(1)} D(\rho)$$

**Proposition 7.3.3**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then there is an exact sequence

$$0 \longrightarrow \mathcal{K}_{X_\Sigma} \longrightarrow \mathcal{O}_{X_\Sigma} \longrightarrow \bigoplus_{\rho \in \Sigma(1)} i_* \mathcal{O}_{D(\rho)}$$

**Proposition 7.3.4**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then  $K_{X_\Sigma}$  is Cartier if and only if there exists  $\{m_\sigma \in \mathbb{Z} \mid \sigma \in \Sigma_{\max}\}$  such that

$$\langle m_\sigma, u_\rho \rangle = 1$$

for all  $\rho \in \sigma(1)$ .

**Proposition 7.3.5**

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then the following are true.

- If  $X_\Sigma$  is projective and Fano, then  $P_{-K_{X_\Sigma}}$  is a reflexive polytope.
- If  $P$  is a reflexive polytope and  $\Sigma = \Sigma_P$ , then  $X_\Sigma$  is Fano.

## 8 Sheaf Cohomology

For a toric variety  $X_\Sigma$  and a divisor  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho)$ , the canonical affine open cover  $\mathcal{U} = \{U_\sigma \mid \sigma \in \Sigma\}$  gives the Cech complex

$$\begin{aligned} C^k(X_\Sigma, \mathcal{U}, \mathcal{O}_{X_\Sigma}(D)) &= \bigoplus_{\sigma_0, \dots, \sigma_k \in \Sigma} \Gamma(U_{\sigma_0} \cap \dots \cap U_{\sigma_k}, \mathcal{O}_{X_\Sigma}(D)) \\ &= \bigoplus_{\sigma_0, \dots, \sigma_k \in \Sigma} H^0(U_{\sigma_0} \cap \dots \cap U_{\sigma_k}, \mathcal{O}_{X_\Sigma}(D)) \\ &= \bigoplus_{\sigma_0, \dots, \sigma_k \in \Sigma} H^0(U_{\sigma_0} \cap \dots \cap U_{\sigma_k}, \mathcal{O}_{X_\Sigma}(D))_\chi \end{aligned}$$

where

$$H^0(U_{\sigma_0} \cap \dots \cap U_{\sigma_k}, \mathcal{O}_{X_\Sigma}(D))_\chi = \begin{cases} \mathbb{C} \cdot \chi & \text{if } \langle \chi, u_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \sigma(1) \\ 0 & \text{otherwise} \end{cases}$$

The Cech complex and hence sheaf cohomology inherits a grading.

Explicitly, we know that  $T$  acts on each affine piece and hence it acts on  $\Gamma(U_\sigma, \mathcal{O}_{X_\Sigma}(D))$  by  $(t \cdot f)(x) = f(t^{-1} \cdot x)$ . Then properties of  $T$  implies that  $T$ -vector spaces are simultaneously diagonalizable, thus we have a decomposition of the Cech complex (either view  $C^k$  as a  $T$ -vector space or see each piece in the direct sum as a  $T$ -space). Then sheaf cohomology also acquires a grading.

### Definition 8.0.1 (The Weight Spaces of Sheaf Cohomology)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho)$  be a divisor. Define the weight spaces of  $D$  to be the components in the direct sum

$$H^k(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) = \bigoplus_{\chi \in \chi(T)} H^k(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))_\chi$$

### Proposition 8.0.2

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Let  $D = \sum_{\rho \in \Sigma(1)} a_\rho D(\rho)$  be a divisor. Then we have

$$H^k(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))_\chi \cong \tilde{H}^{k-1} \left( V_{D, \chi} = \bigcup_{\sigma \in \Sigma} \text{Conv}(\{u_\rho \mid \rho \in \sigma(1), \langle \chi, u_\rho \rangle \leq -a_\rho\}), \mathbb{C} \right)$$

## 8.1 Vanishing Theorems I

### Theorem 8.1.1 (The Batyrev-Borisov Vanishing Theorem)

### Proposition 8.1.2

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Then  $X_\Sigma$  is Cohen-Macaulay.

Since  $X$  is normal and Cohen-Macaulay, we must have

$$\omega_X = \mathcal{K}_X$$

i.e. the dualizing sheaf and the canonical sheaf coincides. Then  $X$  is Gorenstein if and only if  $\omega_X = \mathcal{K}_X$  is invertible if and only if  $\mathcal{K}_X$  is Cartier.

Serre duality holds.

### Theorem 8.1.3 (Toric Serre Duality)

## 8.2 Vanishing Theorems II

Classical vanishing theorems give us the following results. We set  $X_\Sigma$  to be smooth projective and  $D$  ample Cartier divisor.

Kodaira vanishing theorem:

$$H^p(X_\Sigma, \mathcal{K}_{X_\Sigma}(D)) = 0 \quad \text{for all } p > 0$$

Nakano vanishing theorem:

$$H^p(X_\Sigma, \Omega_{X_\Sigma}^q(D)) = 0 \quad \text{for all } p + q > n$$

### Theorem 8.2.1 (Bott-Steenbrink-Danilov)

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a fan on  $\lambda(T)$ . Suppose that  $X_\Sigma$  is projective. Let  $D$  be an ample Cartier divisor. Then we have

$$H^p(X_\Sigma, \hat{\Omega}_{X_\Sigma}^q(D)) = 0$$

for all  $p \geq 1$  and  $q \geq 0$ .

### Proposition 8.2.2

Let  $T$  be a torus over  $\mathbb{C}$ . Let  $\Sigma$  be a complete simplicial fan on  $\lambda(T)$ . Then we have

$$H^p(X_\Sigma, \hat{\Omega}_{X_\Sigma}^q) = 0$$

for all  $p \neq q$ .

## 9 Toric Resolutions and Toric Singularities

## 10 The Chow Ring